SHEEHAN'S CONJECTURE FOR FOUR-REGULAR SIMPLE GRAPHS OF BOUNDED TREEWIDTH

ARGUS OPEN RESEARCH PROJECT

Abstract. Sheehan conjectured in 1975 that no finite simple r-regular graph ($r \geq 3$) possesses a unique Hamiltonian cycle. While odd degrees and all degrees $r > 22$ have been excluded, even degrees $r \in \{4, 6, \ldots, 22\}$—and in particular 4-regular simple graphs—remain open. We prove unconditionally that every finite simple 4-regular Hamiltonian graph of treewidth at most four has at least two distinct undirected Hamiltonian cycles, and more generally that no finite simple graph with two nonadjacent degree-two terminals, nonempty degree-four interior, and a width-four decomposition containing both terminals in one bag admits a unique spanning terminal-to-terminal path.

1. Main Mathematical Results

Theorem 1.1 (Exclusion of unique Hamiltonicity at treewidth four)**.** *Every finite simple* 4*-regular Hamiltonian graph* G *with* $\mathrm{tw}(G) \leq 4$ *possesses at least two distinct undirected Hamiltonian cycles.*

Proof outline. Deleting one vertex u_0 of a hypothetical uniquely Hamiltonian 4-regular simple graph of treewidth at most four and splitting a neighbor along the unique cycle reduces the problem to a two-terminal graph H with nonadjacent degree-two terminals u, v in a common bag of size at most five and degree-four interior vertices. Because local connected two-edge switches can be obstructed on small bags, we pass to a pointed live-projected routing signature algebra on boundaries of size at most five that tracks boundary degree profiles, present boundary edges, and path-pairing multiplicities truncated at two. Exact closure of the pointed support family under introduce, forget, edge-insertion, and join operations shows that every signature reaching a completed terminal state forces path multiplicity at least two. □

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 3167 in the domain of structural graph theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Every finite simple 4-regular Hamiltonian graph of treewidth at most four has at least two distinct undirected Hamiltonian cycles.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted Sheehan's conjecture for 4-regular simple graphs of arbitrary treewidth

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 3167 (structural graph theory), archived in the accompanying repository package `results/3167/`.

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